EAS 5850 “Imaging Informatics”

Summer 2026

Instructors

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Course Description

This 0.5 CU course provides a comprehensive introduction to the field of imaging informatics, with a focus on radiology as the clinical imaging domain. Students will learn about the importance of informatics to the clinical practice of radiology, the unique types of data encountered, relevant data and transactional standards, the growing role of artificial intelligence in radiology, and the challenges faced by imaging informaticists around the globe. This course is geared to any student interested in imaging informatics, and does not require prior training or experience in medicine or medical imaging. Homework assignments include synthesizing reading content and preparing written responses, managing radiology data through coding, and using generative AI to explore health literacy. Unlike other offerings in the course catalog, Imaging Informatics provides a distinctive blend of informatics and radiology, focusing on practical applications and hands-on experience in managing and interpreting medical imaging data, with less focus on intensive coding and technical skill development.

Course Learning Objectives

- Understand the fundamental concepts of imaging informatics

- Compare the similarities and differences between imaging informatics and clinical informatics
- Apply principles of enterprise imaging and artificial intelligence to tackle real-world problems
- Analyze sample data from radiology to understand common imaging informatics tasks and challenges

Course Prerequisite

CIT 5910 “Introduction to Software Development”

Course Textbook

Practical Imaging Informatics, 2nd Edition - available via Penn Libraries
https://link.springer.com/book/10.1007/978-1-0716-1756-4

Grading & Assessment

You must attempt all graded assignments to pass the course. If you have any questions or concerns about grading or progress in the course, please reach out to your instructor.

This course will use a variety of assessments to determine whether learners understand and can apply the key concepts and skills that the course teaches. This includes:

Type	%	Description
Quizzes	5%	Weekly multiple-choice quizzes to test your understanding of the concepts.
Coding Homeworks	45%	There are three coding homeworks in this course. Python will be used for completing them, and the submission will be via Gradescope. Please note, although HW5 is a coding homework that’s partially autograded, the main part of the grade will be based on your approach to solving the problem and the logic that you used, **not** passing the autograder perfectly.
Non-coding Homeworks	30%	There are three non-coding homeworks in this course. HW1 is a quiz submitted via Gradescope, HW4 and HW6 will require paper submission via Canvas.

		There is no homework in Module 7.
Timed Exam	20%	There is one timed exam in Module 7 in the form of a quiz, submitted via Gradescope. The Timed Exam will be cumulative, and will use live online proctoring via Honorlock.
Additional Assignments	0%	There is a non-coding practice assignment before HW3.

Please read the instructions for each assignment very carefully to make sure you know where to submit to receive credit!

Grading / Curving: EAS 5850 faculty reserve the right to apply a curve to final letter grades. None of the assignments or exams will be curved. The following classic numerical ranges will be used to assign your final letter grade:

97 ≤ A+ ≤ 100
93 ≤ A < 97
90 ≤ A- < 93
87 ≤ B+ < 90
83 ≤ B < 87
80 ≤ B- < 83
77 ≤ C+ < 80
73 ≤ C < 77
70 ≤ C- < 73
67 ≤ D+ < 70
60 ≤ D < 67
F < 60

Late Policy

The instruction staff is committed to your success and understands how challenging it can be to learn online while balancing other commitments. Despite your best intentions, sometimes life gets in the way and a little extra time to complete an assignment may be necessary. If you need extra time on an assignment, you can obtain an extension for extenuating circumstances. If an extension is **not** approved, an assignment that is turned in late will receive a late penalty of 10% of the total possible points **per day** up to 5 days. After the 5th day, no credit will be given. Please request extensions via the Extension Request Form located on Canvas Home Page. These extension requests must be submitted **at least 24 hours** before the assignment deadline.

Regrade Requests

Regrade requests are handled on a case-by-case basis and are allowed **up to 1 week** after the grades are released. Requests must be created through the “Regrade Request” function via Gradescope. Regrade requests may take up to a week to process at the discretion of the faculty. When submitting a regrade request, please explain (in detail) why you feel the grading is incorrect.

Extra Credit

There are no extra credit opportunities in this course.

Other Course Activities

The following activities are not mandatory, but will greatly support your success on the graded assignments.

Discussion Forum

Discussion forums (on Ed Discussion) are designed to give you optional extra practice with the material and to see examples of how your classmates are thinking and working. In addition, you can use the discussion forums to ask questions about upcoming assignments and logistical concerns. Course staff will make their best effort to reply within 24 hours to these posts. **Effectively monitoring Ed Discussion for staff announcements, assignment updates, and adjustments to the course schedule is essential for being successful in this course.**

Ed Discussion allows for both public and private posting of questions, comments, and concerns. Private posts should be used anytime you need to discuss sensitive information such as grades, personal issues, or solutions to assignments or exams. Due to the volume of Ed Discussion posts received on a daily basis, our course staff prioritizes the responses to public post as well as private posts containing sensitive information. It is highly encouraged that you create a public post when discussing information that could benefit a wider audience such as assignment and course logistics, or content-related questions. If you wish to create a public post without having your name attached to it you can create a public post anonymously. Please be aware, course staff reserves the right to flip private posts to public and change your identity to anonymous if the post does not contain sensitive information and our course staff believes the content of your post could be beneficial to other students.

Recitation

Recitations are weekly live sessions with your TAs designed around some kind of problem / activity that is supposed to take about an hour to solve. TAs will answer questions as you work through the problem or answer questions that you have submitted in advance of the session. If you are not able to attend either of the scheduled recitation times, you can review a recording of the session, posted the day after.

Additional Segments

The professor may add additional optional segments to support the class as needed.

Creating a Welcoming Environment

All members of the course community – the instructor, TAs, and students – are expected to work together to create a supportive, inclusive environment that welcomes all students, regardless of their race, color, sex, sexual orientation, religion, creed, national origin (including shared ancestry or ethnic characteristics), citizenship status, age, disability, veteran status or any other class protected under applicable federal, state, or local law. **All participants in this course deserve to and should expect to be treated with respect by other members of the community.**

Discussion boards, messaging channels, recitations, office hours, and group working time should be spaces where everyone feels welcome and safe. In order to facilitate a welcoming environment, students of this course are expected to:

- Exercise consideration and respect in their speech and actions
- Attempt collaboration and consideration, including listening to opposing perspectives and authentically and respectfully raising concerns, before conflict
- Refrain from demeaning, discriminatory, or harassing behavior and speech

All members of the course community are expected to be familiar with and abide by the University's guidelines on general conduct and sexual harassment:

- University Code of Conduct: https://catalog.upenn.edu/pennbook/code-of-student-conduct/
- University Sexual Misconduct Policy: https://catalog.upenn.edu/pennbook/sexual-misconduct-resource-offices-complaint-procedures/

Students should also be familiar with other University guidelines regarding personal conduct:

- Conduct & Personal Responsibility guidelines in Pennbook: https://catalog.upenn.edu/pennbook/#policiesbytopictext
- University Principles of Responsible Conduct: http://www.upenn.edu/audit/oacp_principles.htm

If you are a victim of, witness, or are otherwise affected by unacceptable behavior:
- In cases of sexual harassment or assault, please consult DPS Special Services (https://www.publicsafety.upenn.edu/about/special-services/sensitive-crimes/ at 215-573-3333; this is a confidential resource.
- To report other bias incidents, please complete the Bias Incident Reporting Form: https://belonging.upenn.edu/diversity-at-penn/bias-motivated-incident-report
- For other violations of the code of student conduct, the Center for Community Standards and Accountability has an incident reporting form at https://csa.upenn.edu/community-standards/refer-caserequest-consultation

If you are unsure which office to contact, please contact the instructor or any Penn Engineering Online Learning staff member.

Getting and Giving Help

TA and Faculty Support

TAs will hold office hours weekly when they will offer 15-minute time slots on a first come first serve basis. Your professor will hold weekly open office hours and be available for a limited number of private meetings per semester, depending on the needs of the class.

Collaboration Guidelines

In the professional world of software development, collaboration—including using code that others have written—is both practical and ubiquitous.

However, to prepare for entering that professional context, you need to develop a full set of software development skills so that you are both able to create your own code and evaluate the quality of someone else's code that you might use.
In the context of this course, independent work and evaluation is critical. **Do not collaborate with others on individual graded assignments unless it is explicitly indicated.** Inappropriate collaboration will be considered cheating and considered under Penn's Code of Academic Integrity.

Please note that searching for solutions or code online is a violation of academic integrity. Sharing solutions or code with another student (unless working on a group project or other collaborative assignment) is also a violation of academic integrity. This includes posting solutions and code publicly online, even after you've completed the course. If you discover publicly viewable solutions for the assignments of this course, please let the course staff know immediately.

Discussion forums and recitations *are* collaborative—please take advantage of those times to work with your colleagues. For general communication with your colleagues, use your Slack channels or Slack direct messages.

Guidelines for the Use of Generative AI in this Course

We recognize the increasing prevalence and power of Artificial Intelligence (AI) tools, and encourage their responsible and ethical use to enhance your learning experience. However, it's essential to develop a strong understanding of fundamental processes before relying on AI. To that end:

Examples of acceptable uses of AI

- Comprehension and Expansion: You are encouraged to use AI to clarify and expand your understanding of course materials.
- Research and Information Gathering: AI can supplement your research and information gathering, aiding your exploration of complex concepts.

Examples of unacceptable uses of AI

- Assignment Completion: Avoid using AI for generating code or text for assignments, peer reviews, or content summaries, *unless explicitly instructed by the course staff.*
- Assessment Assistance: Do not seek AI help for quizzes, exams, or assessments.
- Academic Integrity: Do not present AI-generated content without citation and as your own work.

Engaging in unacceptable use of AI tools will result in academic consequences, which may include grade penalties, academic warnings, or other actions as determined by your instructor and the university's academic integrity policies.

Note that these guidelines may differ from those in other courses. If you have questions or concerns, don't hesitate to reach out. Ultimately, AI can be a valuable educational tool when used responsibly and aligned with our course policies.

Jean

Our program's new AI chatbot, Jean, will be piloted in this course's Ed Discussion. For all up-to-date guidance on Jean, please see the student guidelines found within the Canvas course.

Plagiarism Policy

Students found plagiarizing on any assignment will automatically receive a zero on that assignment. The first offense of plagiarism on an assignment will be handled by the instructor. Only in extenuating circumstances, or if the plagiarism was committed on an exam, will a first offense be turned over to the University of Pennsylvania Center for Community Standards and Accountability.

A second offense, or, exam plagiarism will be turned over immediately to the University of Pennsylvania Center for Community Standards and Accountability.

A report being submitted to the University of Pennsylvania Center for Community Standards and Accountability will result in a failure of the course (regardless of current course average), and a potential permanent notation on your academic record that will follow you to all future academic institutions and possibly future employers. If you are unfamiliar with what constitutes plagiarism at Penn, visit Penn's Code of Academic Integrity.

Recording Notice

Public office hours, recitations, and other live events will be recorded, used, and may be made available to class participants during the current semester as well as students who take the class in future semesters.

Private office hours will also be offered and are not recorded. Students who do not wish to attend the publicly-recorded office hours may attend the private office hours.

Access to Materials and Content Before and After Graduation

If you would like to retain copies of your submitted assignments, you must download them from Gradescope, Canvas, Codio, and any other platforms that you submit to during the semester in which you are taking that course.

Access to course materials and your submissions is not guaranteed after the completion of a course. Therefore, we recommend that students download any assignments or materials they would like to keep before a course concludes.

Summer 2026 Course Schedule and Important Dates

Dates are subject to change. Please check Ed Discussion for announcements regarding schedule changes. Note, weeks run Monday through Sunday.

Monday, 5/4 - Sunday, 5/10	**Module 1**	**Topic(s)** Intro to Informatics and Working with Healthcare Data	**Assignment(s)** HW1: Getting Familiar With Imaging Informatics M1: Quiz
Monday, 5/11 - Sunday, 5/17	**Module 2**	**Topic(s)** Role of Standards and Ontologies	**Assignment(s)** HW2: De-identify a DICOM Image M2: Quiz
Monday 5/18 - Sunday 5/24	**Module 3**	**Topic(s)** Clinical Workflow, Tools, and Standards	**Assignment(s)** HW3 Practice Assignment HW3: Analyze an HL7 Message M3: Quiz
Tuesday 5/26 - Sunday 5/31 Memorial Day 5/25 - No Class	**Module 4**	**Topic(s)** Enterprise Imaging Informatics	**Assignment(s)** HW4: Rethinking Enterprise Imaging From the Ground Up M4: Quiz
Monday 6/1 - Sunday 6/7	**Module 5**	**Topic(s)** Artificial Intelligence	**Assignment(s)** HW5: Extract Recommendations From Radiology Reports With NLP M5: Quiz
Monday, 6/8 - Sunday, 6/14	**Module 6**	**Topic(s)** Challenges and Opportunities	**Assignment(s)** HW6: Use ChatGPT to Explore Health Literacy M6: Quiz
Monday, 6/15 - Sunday, 6/21 Juneteenth 6/19 - No Class	**Module 7**	**Topic(s)** Informatics for Global Health (Guest Faculty: Farouk Dako, MD)	**Assignment(s)** Timed Exam (Proctored by Honorlock)